Region vs. Genre Video Game Sales Efficiency using Clustering Algorithms

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Introduction

The video game development process is different from other creative media. A genre for a game is the foundation, changing a genre mid-development can completely set back a team to their beginning development phase. Our goal of this project was to design an unsupervised machine learning algorithm to help developers in making their early important decision as well as support the developer's decision when presenting to possible investors. This project will show the relationships between genres and region and how well they have sold, giving the user the best options as result.

Related Works

Akinwale Mutiat, Analysis of Video Game Sales Across Regions Of The World
https://mutiatakinwale.medium.com/an-analysis-of-video-game-sales-across-regions-of-the-world-31c2215d4cc9

Ricardo C. Muller, Video Game Sales Analysis and Clustering
https://github.com/ricardocmuller/Video_Game_Sales_Analysis_and_Clustering/tree/main

Video Game Sales dataset Discussion Board
https://www.kaggle.com/datasets/gregorut/videogamesales/discussion/455902

Dataset

Source: Kaggle - Video Game Sales by Gregory Smith (https://www.kaggle.com/datasets/gregorut/videogamesales)

Type: Structured

Instances: 11493 values

Features: Rank, Name, Platform, Year, Genre, Publisher, NA_Sales, EU_Sales, JP_Sales, Other_Sales

Format: .csv

Pre-processing: Split up dataset into genre-level profiles, made them into two with Principal Component Analysis (PCA)

Unsupervised Algorithms

-K-Means - Baseline Model

-Agglomerative Hierarchy - Take General Clusters made by Baseline and

- Gaussian Mixture Model - Use to compare with Results

Visualizing Clustering

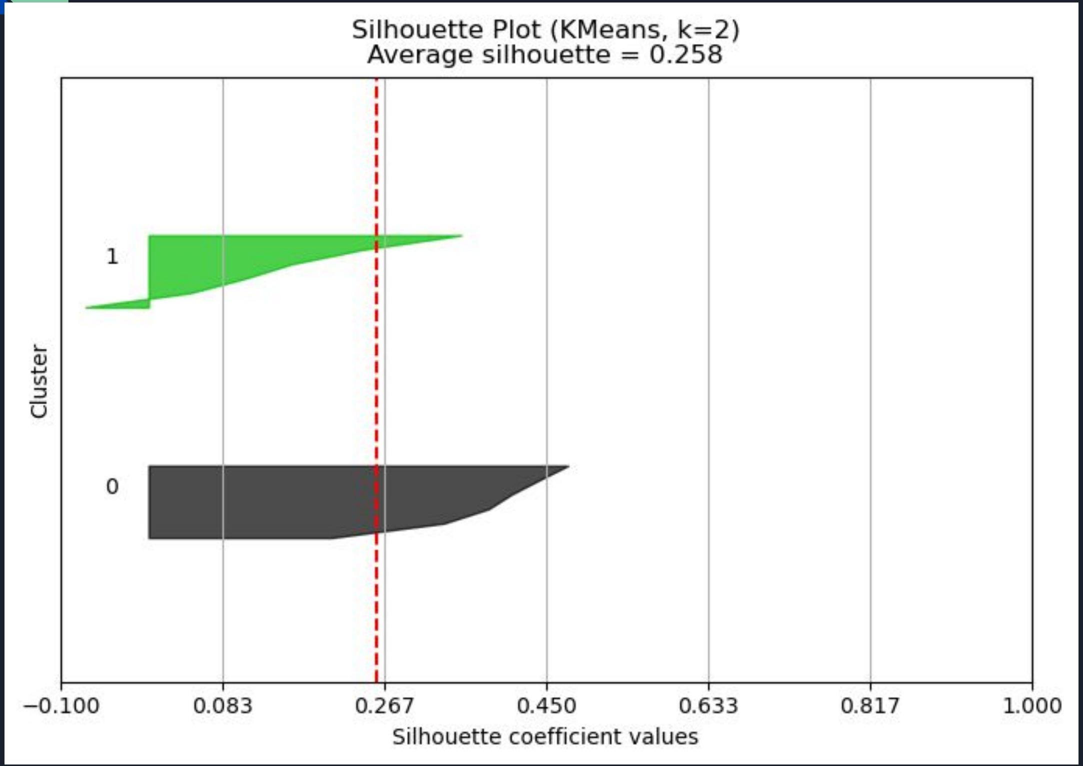
Scatter plots

Dendrogram

Silhouette plots

Principal Components Analysis (PCA)

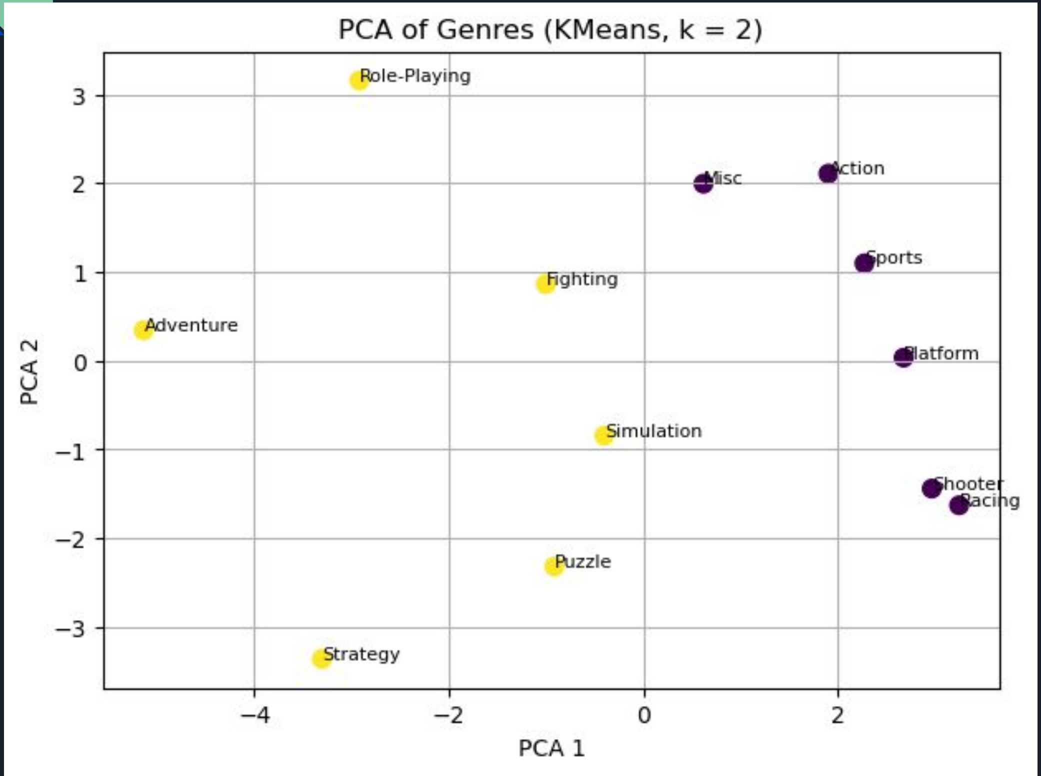
Results



- We used Sihouette_score to determine our best K

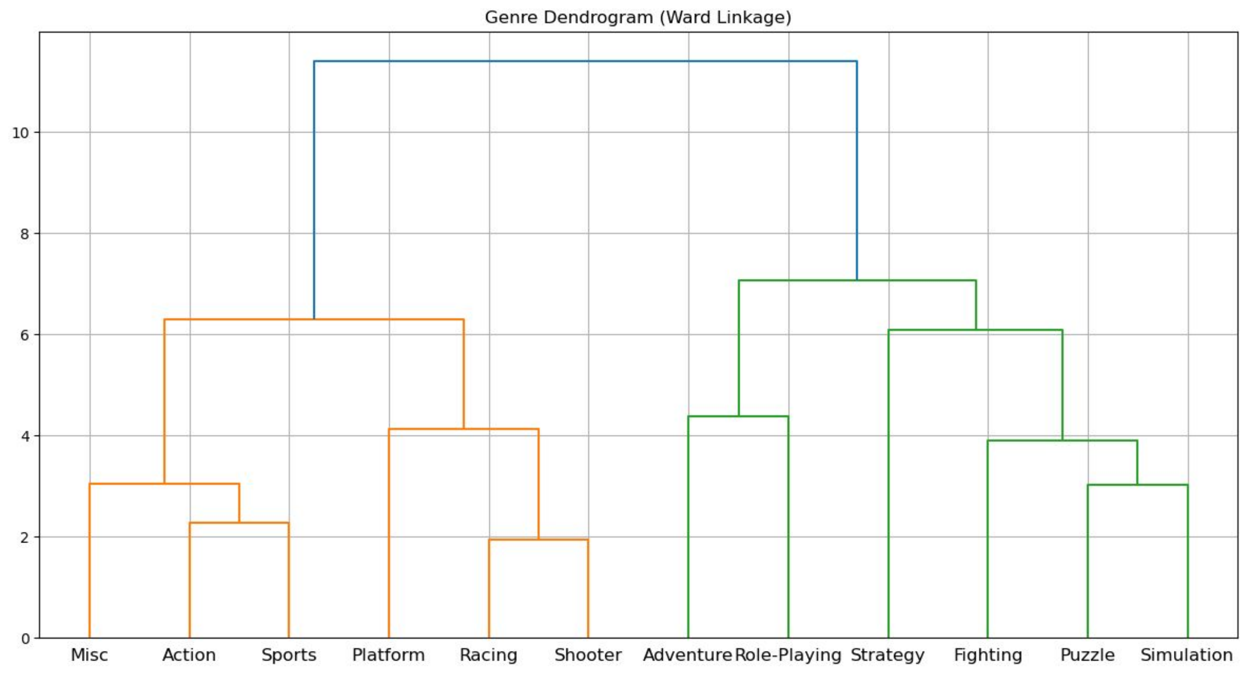
- Tested all possible clusters and select K with best Silhouette Score

Results



- Closer dots = Closer Relationship

Results



- Closer genres = similar relationship

Discussion & Analysis

- Genre → market share, total sales, mean release year, entropy of regional dominance
- PCA captures 75% total variance
- Silhouette score → k = 2 → optimal clusters
- RPG > in Japan; Action > everywhere else
- When k = 6, clusters group and show niches
- All algorithms were consistent with one another
- Dataset last updated 7 years ago
- Use more recent dataset with less ambiguous genres

Teamwork and Task Division

David Forero -

Visualization of Algorithms, Debugging, Dataset Acquisition, Report Work, Slideshow Preparation

Robert Castanon -

Clustering Algorithms, Debugging, Dataset Formatting, Report Work, Slideshow Preparation

Project timeline

WEEK 8 — Reconfirm Group and Define Real-World problems

WEEK 9 — Finalize Dataset, Goal, and Research algorithms

WEEK 10 — Finalize methods and code development

WEEK 11 — Code Debugging

Lessons Learned

- Importance data processing and its effects on clustering algorithms.
- How to determine amount of clusters
- How number of clusters affect cluster output.
- How to properly analyse data on graphs.

Code Presentation

Thank you!